Yusuf Tahir

Undergraduate Researcher | Computer Science & Mathematics

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Summary

Undergraduate researcher double-majoring in Computer Science and Mathematics at Krea University, working at the intersection of quantum computing, machine learning theory, and mathematical optimization. My research combines theoretical analysis, mathematical proof techniques, and computational simulation to study quantum routing, amplitude amplification, and variational quantum algorithms. I contribute to research manuscripts and open-source quantum software projects, with a focus on translating mathematical ideas into practical computational systems.

Research Interests

Amplitude Amplification, Computer Vision, Machine Learning Theory, Mathematical Oracles, Natural Language Processing, Quantum Machine Learning, and Quantum Variational Algorithms.

Selected Coursework (Independent Learning)

- **Computer Science:** Machine Learning, Cryptography, Data Structure & Algorithms, Optimization Methods, Advanced NLP, Theory of Computation, Deep Learning, Advanced/Quantum Complexity Theory
- **Mathematics & Physics:** Inference and Information, Statistics, Discrete Applied Mathematics, Abstract Algebra, Real & Complex Analysis, Quantum Mechanics, Statistical Physics, Electromagnetism, General Relativity, Quantum Field Theory, Quantum Information

Education

KREA UNIVERSITY 2025–2029 (Expected)
Double Major: Computer Science and Mathematics — GPA: 9.13
Additional Interest: Physics (independent coursework and auditing)

AL-MOMIN INTERNATIONAL SCHOOL Grade 12 2023–2024
CBSE Board — Percentage: 74%
Relevant Subjects: Mathematics, Physics, Chemistry, English, Physical Education

AL-MOMIN INTERNATIONAL SCHOOL Grade 11 2022–2023
CBSE Board — Percentage: 90.6%
Relevant Subjects: Mathematics, Physics, Chemistry, English, Physical Education

A S S PATNA CENTRAL SCHOOL Grade 10 2012–2022
CBSE Board — Percentage: 93.4%
Relevant Subjects: Science, Mathematics, Social Studies, English, Information Technology, Hindi

Work Experience

Undergraduate Researcher — Krea University (Prof. Mithilesh Kumar) Feb 2026 – Present

- Conduct research in quantum algorithms, quantum routing, amplitude amplification, and variational quantum computing.
- Formulate theoretical frameworks, derive mathematical proofs, perform analytical calculations, and design computational simulations.
- Develop and validate implementations using Cirq, Qiskit, and QuTiP.
- Co-author multiple research manuscripts and open-source quantum software libraries, including contributions as an equal author on several projects.

Preprints and Publications

Dealing with locality in QAOA Jun 2026
Kumar, M.*, **Tahir, Y.*** (*Equal contribution)
Submitted to Quantum
Preprint available at: DOI: 10.48550/arXiv.2606.14447 | arXiv: 2606.14447

- Proposed a transport-augmented QAOA enriching the mixer with optimized shortcut couplings to collapse effective interaction-graph diameter.
- Connected optimal shortcut placement to bounded-diameter graph augmentation via exact finite-depth support recursions.
- Demonstrated size-invariant performance improvement for MaxCut on bipartite families and random trees at shallow depths.

Congestion-free routing on quantum chips Apr 2026
Kumar, M., **Tahir, Y.**, Daiya, V., Mattaparthi, S., Shaurya, A.
Minor revisions requested at Physical Review A (second review ongoing)
Preprint available at: DOI: 10.48550/arXiv.2604.27015 | arXiv: 2604.27015

- Designed a swap-free routing framework using orthogonal spectral buses to transport control information without moving the computational state.
- Proved decodability, reversibility, and correctness for CNOT and Boolean fan-in operations, supported by Cirq and QuTiP simulations.
- **Contribution:** Led the research effort following the initial concept proposed by Prof. M. Kumar, including formulation of the theoretical framework, derivation of mathematical proofs, analytical calculations, simulation development in Cirq and QuTiP, and preparation of the manuscript and appendices.

`ampamp`: A Comprehensive Python Library for Quantum Amplitude Amplification and QSVT May 2026
Kumar, M.*, Daiya, V.*, **Tahir, Y.*** (*Equal contribution)
Submitted to SciPost Physics Codebases
Links: GitHub Repository

- Presented a Python library for quantum amplitude amplification, QSVT, and circuit-analysis workflows.
- Provided engines for Grover search, fixed-point and oblivious amplification, distributed partitioning, variable-time models, and polynomial synthesis.
- Detailed public classes, inputs, outputs, and examples for clear, reusable, practical quantum-algorithm engineering studies.

Amplitude Amplification Algorithms May 2026

Kumar, M.*, **Tahir, Y.***, Daiya, V.* (*Equal contribution)

(Editorial inquiry submitted to PRX Quantum and npj Quantum Information)

Preprint available at: DOI: 10.5281/zenodo.20054981

- Formulated a unified mathematical formalism tracing quantum amplitude amplification from Grover's algorithm to the Quantum Singular Value Transformation (QSVT) framework.
- Introduced a variational template for amplitude amplification based on parameterized phase rotations.
- Created a companion website hosting interactive visualizations and implementation details.

Projects & Initiatives

`QCore`**: Quantum-Classical Compiler and Runtime Engine** Apr 2026 – Present

- Independently developing a backend-agnostic quantum compiler and runtime engine that replaces Python-orchestrated workflows with a compiler-oriented execution system.
- Built a Rust-based frontend/compiler prototype featuring real intermediate representation (IR), kernel decomposition, replay tracing, and backend parity checks.
- Implemented a verification-heavy, CPU-first execution pipeline with a GPU stub backend for contract validation.

`ampamp`**: Amplitude Amplification Tooling for Python** Feb 2026 – Present

- Co-authored a Python library (with V. Daiya) to build, inspect, profile, and validate quantum amplification workflows.
- Packaged Grover search, fixed-point schedules, oracle construction, variable-time models, and QSVT/QSP helpers.
- Integrated diagnostics, transpilation profiling, and backend validation behind a compact research API.

RateMyProf: Student Professor Review Platform Jan 2026 – Present

- Developing a full-stack web application (Next.js, Express.js, Supabase) for Krea University to provide centralized, searchable, and anonymous academic reviews.
- Implementing real-time aggregated analytics, secure Google OAuth authentication, role-based access control, and an administrative moderation panel.
- Designed to support informed course selection; currently in active development and targeted for campus-wide deployment in the next academic year.

Krea Parcel Scanner System Jan 2026 – Present

- Digitizing campus security workflows by engineering a real-time parcel management system using Firebase Firestore.
- Integrating QR code scanning, OCR-based name extraction, and automated weekly CSV archival across Dispatch and Delivery interfaces.
- Designed to reduce manual workload and streamline parcel processing; under administrative review for campus deployment.

Strategic Development Proposal for Krea University Jan 2026 – Present

- Authored an independent strategic roadmap outlining a multi-pronged framework for institutional development, focusing on academic excellence, research culture, industry integration, and visibility.
- Independently submitted the proposal to university leadership, resulting in detailed written feedback from the Vice Chancellor and active follow-up correspondence on institutional reform strategies.

Technical Skills

- **Programming:** C++ (advanced), Python (advanced), Rust (proficient)
- **Quantum:** Cirq, Qiskit, QuTiP
- **Libraries/Frameworks:** NumPy, Pandas, Matplotlib, PyTorch, TensorFlow
- **Tools:** Git, Linux, terminal workflows
- **Languages:** English, Hindi, Urdu (native)

Extracurricular Activities

Volunteer Tutor for Underprivileged Children 2024

- Taught mathematics and science to 15–20 Grade 5 students from nearby underserved communities.
- Provided basic education support and motivation to continue learning

Entrepreneurial Experience 2023 – Present

- Contributed to two early-stage startup projects involving merchandising and computational skincare products.
- Assisted in planning, outreach, web design, finances and early operations

Awards & Achievements

- School Topper – Grade 10
- Best Student Award (Awarded for overall excellence in academics and conduct)